RIVER VALLEY HIGH SCHOOL

YEAR 6 PRELIMINARY EXAMINATION

CANDIDATE NAME	
CLASS	6
CENTRE NUMBER	S
INDEX NUMBER	

H1 CHEMISTRY **8872/02**

Paper 2

11th September 2012

2 hours

Candidates answer Section A on the Question Paper.

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name, class, Centre number and index number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A
Answer all the questions.

Section B
Answer two questions on separate answer paper.

A Data Booklet is provided. Do not write anything on it.

The number of marks is given in brackets [ ] at the end of each question or part question.
At the end of the examination, fasten all your work securely together.

For Examiner's Use	
Section A	
B5	
B6	
B7	
Total	

This document consists of **12** printed pages.

1. Below is a list of elements from the third period:

Period 3	Na	Mg	A*l*	Si	P	S	C*l*

(a) Which of these element(s)

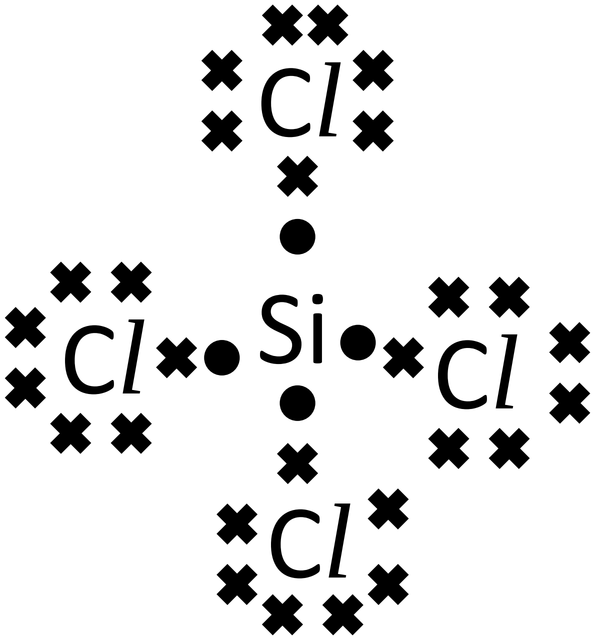
(i) have a giant metallic structure

Na, Mg and A*l*

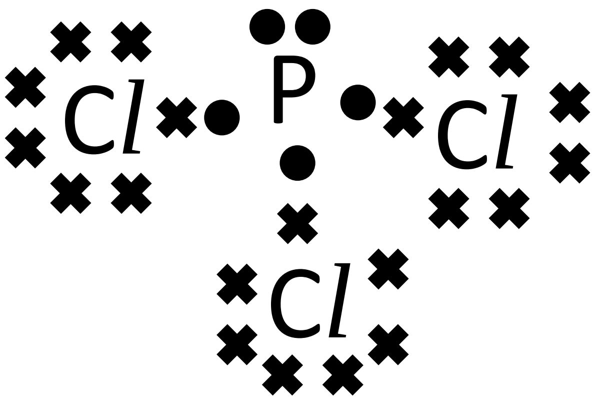
(ii) have a giant covalent structure **[2]**

Si

(b) Both Phosphorous and Silicon can react with chlorine to form PCl_3 and $SiCl_4$ respectively. Draw the dot-and-cross diagram for both compounds, state the shape and suggest the values of the bond angles **[3]**



Tetrahedral, 109.5°



Trigonal pyramidal, 107° or < 109.5°

[Total : 5]

2. Sulfuric acid is commonly used in various industries as a feedstock for the production of other useful chemicals such as fertilizers, synthetic resin, detergents and insecticide.

(a) The contact process is the current method for producing sulfuric acid for use in various industrial processes. The main reaction of the contact process is shown:

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ $\Delta H = -197 \text{ kJ mol}^{-1}$

An **<u>equimolar</u>** mix of SO_2 and O_2 is introduced into a 1 dm^3 sealed flask.

At equilibrium, 33.3% of O_2 is used up and the total amount of gas $(n_{SO_2} + n_{O_2} + n_{SO_3})$ equals to 0.241 mol

(i) Write down an expression for the K_c of the reaction, stating the units.

$$K_c = \frac{[SO_3]^2}{[SO_2]^2[O_2]} \text{ mol}^{-1}\text{ dm}^3$$

(ii) Calculate the value of K_c **[4]**

Let the initial $[O_2]$ be x:

	$2SO_2(g)$	+	$O_2(g)$	$\rightleftharpoons$	$2SO_3(g)$
Initial concentration/ $moldm^{-3}$	x		x		0
Change in concentration/ $moldm^{-3}$	$-0.667x$		$-0.333x$		$+0.667x$
final concentration/ $moldm^{-3}$	$0.333x$		$0.667x$		$0.667x$

$1.667x = 0.241 mol$

$x = 0.145 \text{ } mol$

$$K_c = \frac{[SO_3]^2}{[SO_2]^2[O_2]}$$

$$K_c = \frac{\left[\frac{2(0.145)}{3}\right]^2}{\left[\frac{0.145}{3}\right]^2\left[\frac{2(0.145)}{3}\right]}$$

$K_c = 41.4 \text{ mol}^{-1} \text{ dm}^3$

(b) Unlike sulfuric acid, ethanoic acid, CH_3COOH, is a weak acid.

(i) Explain what does the term 'weak acid' means

It is a substance that dissociate partially when dissolved in water to give H^+/ H_3O^+ ions.

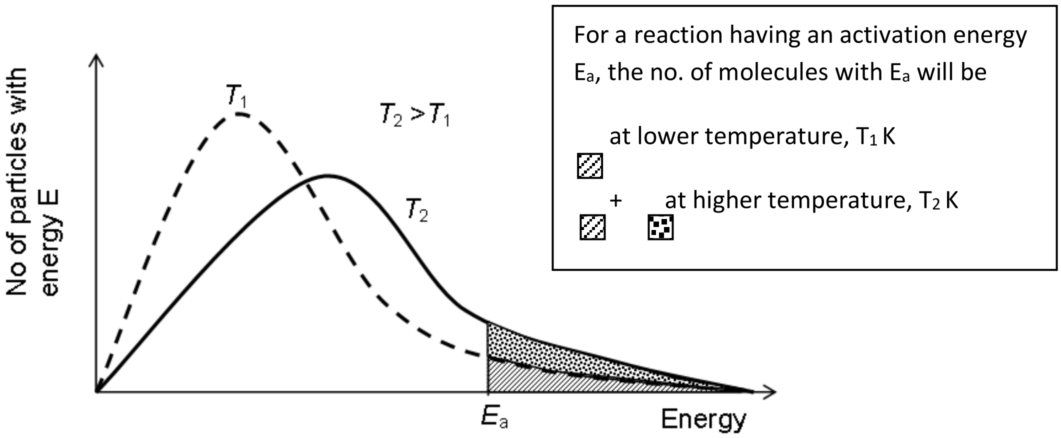
(ii) Write down the expression for K_a of ethanoic acid **[2]**

$$K_a = \frac{[CH_3COO^-][H^+]}{[CH_3COOH]}$$

[Total : 6]

3. The rate of reaction is affected by factors such as temperature and catalyst.

(a) With the help of an appropriate sketch of the Boltzmann distribution, explain how the increase in temperature speeds up a chemical reaction **[3]**



At a higher temperature, there is a larger number of reactant particles with energy greater than or equal to the activation energy**[1m]** resulting in an increase in the frequency of *effective* collisions **[1]** between the reactant particles and hence an increased rate of reaction.

(b) Other than temperature and catalyst, suggest and explain another factor which will affect the rate of reaction. **[2]**

Surface area/concentration

As the concentration of the reactant is increased, the number of reactant particles increases. Consequently, there will be an increase in the frequency of effective collisions.

OR

When the surface area increase, there is a larger surface area of contact between the reactant particles and hence, a higher frequency of effective collisions between the reactant particles.

(c) Aluminium oxide is a common oxide which is insoluble in water and *amphoteric* in nature.

(i) Define the term *amphoteric*

A compound which is amphoteric can react with both acid and base to produce salt and water.

(ii) Describe two reactions which show the amphoteric behaviour of aluminium oxide.

With acid: $Al_2O_3(s) + 6H^+(aq) \rightarrow 2Al^{3+}(aq) + 3H_2O(l)$

With base: $Al_2O_3(s) + 2OH^-(aq) + 3H_2O(l) \rightarrow 2[Al(OH)_4]^-(aq)$

(iii) Explain why aluminium oxide is insoluble in water. **[5]**

$Al_2O_3(s)$ is insoluble in water due to the large magnitude of its lattice energy. The hydration energy released through the formation of ion-dipole interaction is insufficient to compensate for the energy required to break down the giant ionic lattice.

[Total : 10]

4. There are some wines that taste better after a period of 'ageing' process. After ageing, the wine is said to taste more flavourful is more appealing to consumers. One of the many reactions that takes place in the wine is the alcohol (ethanol) reacting with the fruit acids

Reaction (1):

$$RCOOH(aq) + CH_2CH_3OH(aq) \rightleftharpoons RCOOCH_2CH_3(aq) + H_2O(l)$$

Where RCOOH represents the fruit acids. The ageing process usually involves placing the wine bottles in a cold environment (about 13 °C) for periods of about 1 year. In time, the ester that is produced will give the 'flavour' that aged wine has. However, the wine can be easily 'damaged' if the bottles are exposed to high heat, even for short periods of time. When this happens, the wine becomes sour to the taste.

Wine can also be 'damaged' if it is exposed to the air. In this case, the alcohol (ethanol) inside the wine is converted to ethanoic acid by oxygen in the air (where oxygen is converted to water). This event may happen during the bottling stage, where air is unintentionally introduced into the bottle.

(a) **(i)** *'...placing the wine bottles in a cold environment...'* From this statement, what can you deduce about the enthalpy change of reaction (1)? Explain your answer.

(Forward reaction) of reaction 1 is <u>exothermic</u>. At low temperatures (By Le Chatelier's principle): <u>Equilibrium position shifts to the right/ The exothermic reaction is favoured (to produce heat)</u>

(ii) Hence, explain why "...*the wine can be easily 'damaged' if the bottles are exposed to* ***<u>high heat</u>****, even for* ***<u>short periods of time</u>****... becomes* ***<u>sour</u>*** *to the taste.*", paying particular attention to the underlined phrases. **[5]**

At high temperature, the <u>backward reaction of reaction (1) is favoured</u>. The <u>reaction rate is </u>also <u>faster</u> (due to the high temperature). Hence, the <u>acid concentration/ amount is increased</u> quickly and the wine gives a sour taste.

(b) **(i)** By means of two half equations, write a full balanced equation for the oxidation of ethanol to ethanoic acid.

$CH_3CH_2OH + H_2O \rightarrow CH_3COOH + 4H^+ + 4e^-$

$O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$

$CH_3CH_2OH + H_2O + O_2 \rightarrow CH_3COOH + 2H_2O$

(ii) During a bottling process, 8 cm^3 of air is introduced by accident into a 750 cm^3 wine bottle. What is the maximum concentration (in mol dm^{-3}) of ethanoic acid that can be produced?

Assume that ethanol is in excess and air contains 20% oxygen **[4]**

$CH_3COOH \equiv O_2$

$$n_{O_2} = \frac{(0.2 \times 8)}{24000}$$

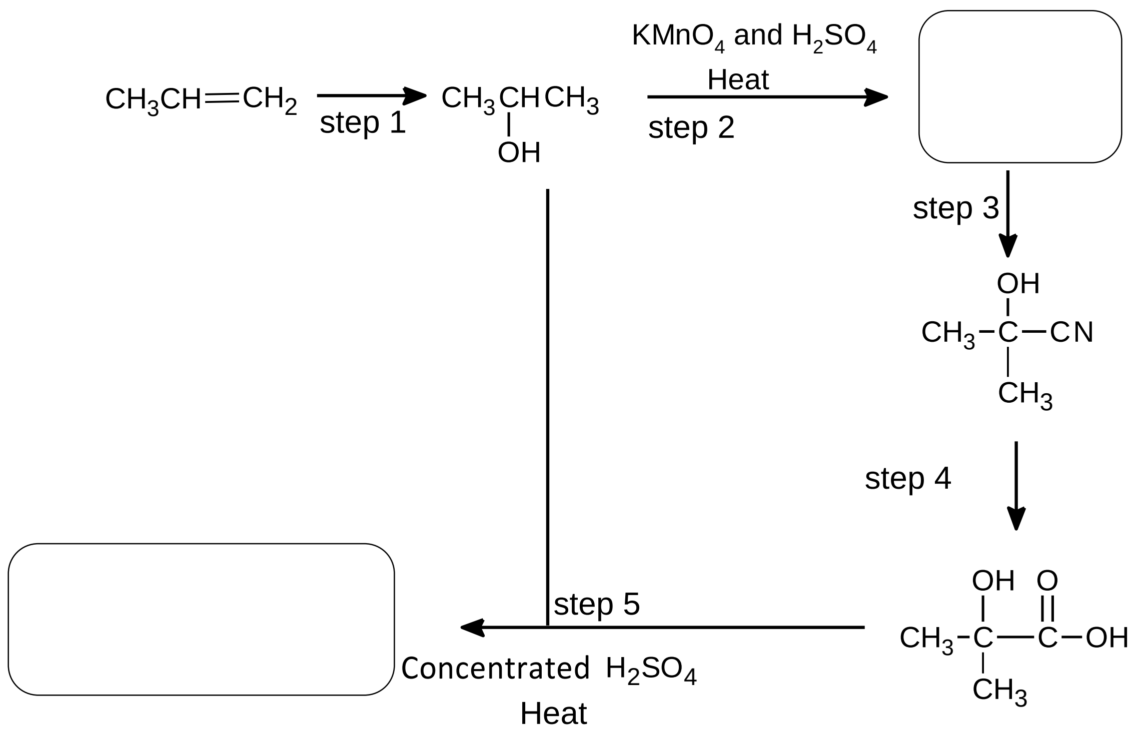
$$n_{O_2} = 6.67 \times 10^{-6} mol$$

$$[CH_3COOH] = \frac{6.67 \times 10^{-6}}{750} \times 1000$$

$$[CH_3COOH] = 8.89 \times 10^{-6} \text{ mol}^{-1} \text{ dm}^3$$

[Total : 9]

5 (a) The diagram below shows the synthesis pathway starting with propene:

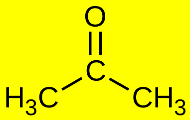


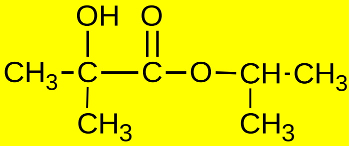
(i) State the reagents and conditions for step 1 and 3

Step 1: Steam, H_3PO_4, 60 atm, 300 °C

Step 3: HCN, trace NaCN / NaOH

(ii) Draw the structure of the products of step 2 and 5

Step 2: 

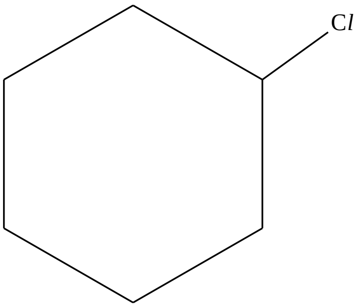
Step 5: 

(iii) Name the type of reaction taking place in step 5: **[5]**

Condensation

(b) Suggest a simple chemical test to distinguish the following pairs of compounds. Your answers should include the reagents and conditions for each test and the observations you would expect to see for each compound

(i)



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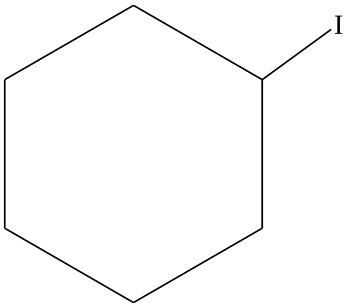
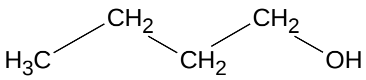


Test: Add NaOH and heat, followed by HNO_3 and $AgNO_3$

Observstion: White ppt formed for $C_6H_{11}Cl$

Yellow ppt formed for $C_6H_{11}I$

(ii)



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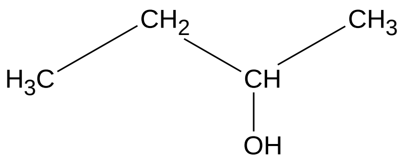


[5]

Test: aq I_2, NaOH, warm

Observation: $CH_3CH_2CH_2(OH)CH_2$ will form a yellow ppt

$CH_3CH_2CH_2CH_2OH$ will not form a yellow ppt.

[Total : 10]

Section B

Answer **two** of the three questions in this section on separate answer paper.

6 Larger alkane molecules in petroleum can be broken down into smaller molecules in a process called cracking. Cracking is carried out on fractions obtained from fractional distillation of petroleum. An example of cracking is decane, $C_{10}H_{22}$. The reaction is shown below.

$$C_{10}H_{22} \rightarrow C_2H_4 + C_4H_{10} + C_3H_8 + C$$

With the production of alkanes and alkenes, these organic compounds have many different uses and can undergo many different organic reactions to form other products.

(a) Two of the products that can be formed from ethene are ethanal and ethanol.

(i) Ethanal can be produced from ethene under laboratory conditions via an organic synthesis. Suggest suitable reagents and conditions for each step(s) to produce ethanal.

Step 1 : HC*l*, heat

Step 2 : aqueous NaOH, heat

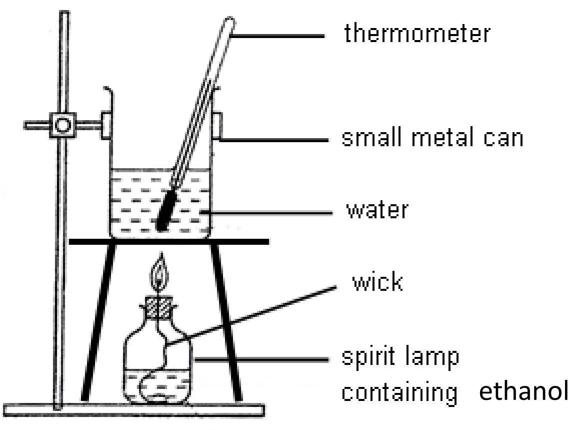
Step 3 : acidified $K_2Cr_2O_7$, distill conditions.

(ii) The standard enthalpy change of combustion of ethanol can be determined by conducting an experiment in the laboratory.

1. Write an equation for the complete combustion of ethanol.

$CH_3CH_2OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l)$

2. With the aid of a diagram, briefly describe how the enthalpy change of combustion of ethanol can be determined using laboratory apparatus.



A known amount of ethanol, eg 0.96g, is burnt.

The temperature rise of the known volume water, eg : 200cm^3, in the metal can is noted.

Enthalpy change of combustion can be calculated.

3. A student conducted the experiment in the laboratory and the data collected are as follows:

Mass of ethanol used	= 0.96g
Volume of water used in heating	= 300 cm^3
Initial temperature of water	= 28.0 °C
Final temperature of water	= 49.6 °C

Using the specific heat capacity value in the data booklet, calculate the standard enthalpy change of combustion of ethane.

Heat absorbed by water = $300 \times 4.18 \times (49.6 - 28.0)$

= 27086 J = 27.1 kJ

Amount of ethanol burnt = $\dfrac{0.96}{12.0x2 + 16.0 + 6.0}$

= 0.0209 mol

ΔH_c (ethanol) = $-\dfrac{27086}{0.0209}$ = -1.30×10^3 kJ mol^{-1}

4. The theoretical value for the enthalpy change of combustion for ethanol is -1370 kJ mol^{-1}. Referring to in **a(i)(3)**, state a reason for the difference obtained.

Energy produced from burning of ethanol is lost to the surrounding.

5. State one use of ethanol. **[12]**

Solvents for paints/ vanishes/ perfumes/ Alcoholic drinks/

Fuel for vehicles

(b) Alkanes produced from cracking are often used as petrol and diesel for motor vehicles. State one example of a pollutant produced from the incomplete combustion of the alkane and its harmful effect on the environment. **[2]**

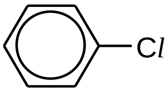
CO : CO contributes to the formation of smog, ground-level ozone, which can trigger serious respiratory problems.

Unburnt hydrocarbon : Forms part of photochemical smog

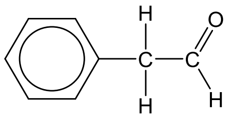
NO, NO_2 : forms acid rain

(c) Distinguish the following pairs of organic compounds. Include reagents, conditions, observations and equations for any reactions in your answers.

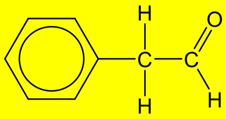
(i)



and



1) 2, 4 DNPH

Orange ppt observed when added to 

Eqn :

(ii) $CH_3COOC(CH_3)_3$ and $CH_3COOCH_2CH_3$ **[6]**

1) Add acidified $K_2Cr_2O_7$ heat

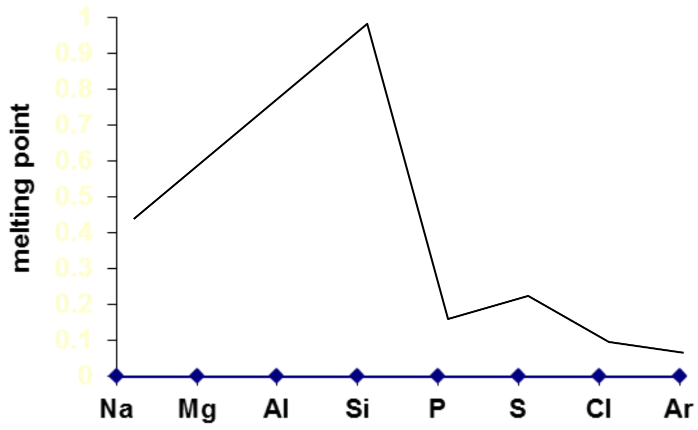
Orange $K_2Cr_2O_7$ turned green observed in solution containing $CH_3COOCH_2CH_3$

$CH_3CH_2OH + 2[O] \rightarrow CH_3COOH + H_2O$

[Total: 20]

7. Sodium, aluminium and phosphorous are elements are some examples of elements in Period three. Their melting point varies across the period depending on their structures.

(a) **(i)** Sketch a diagram to show the variation in melting point of the elements in Period three.



(ii) Explain the diagram drawn in **(a)(i).**

The metals (Na, Mg and Al) have higher melting points as they all have giant metallic structures and a larger amount of energy is required to overcome the strong electrostatic attraction between the positive metal cations and mobile valence electrons.

Si has the highest melting point as it has a giant covalent structure and a large amount of energy is required to overcome the strong covalent bonds between the Si atoms.

The non-metals (P to Ar) have lower melting points as they all have simple covalent/molecular structures or monoatomic structure in the case of Ar. Less energy is required to overcome the weaker van der Waals' forces between the molecules or between the atoms in the case of Ar.

(iii) Explain, with the help of equations, the reaction of chlorides of, aluminium and phosphorous with water. **[7]**

$AlCl_3$ undergoes substantial hydrolysis in water to give an acidic solution with pH = 3.

$AlCl_3(s) + 6H_2O(l) \rightarrow [Al(H_2O)_6]^{3+}(aq) + 3Cl^-(aq)$

$[Al(H_2O)_6]^{3+}(aq) + H_2O(l) \rightleftharpoons [Al(H_2O)_5(OH)]^{2+}(aq) + H_3O^+(aq)$

P_4O_{10} reacts with water to give an acidic solution with pH = 2.

$PCl_5(s) + 4H_2O(l) \rightarrow H_3PO_4(aq) + 5HCl(aq)$

OR

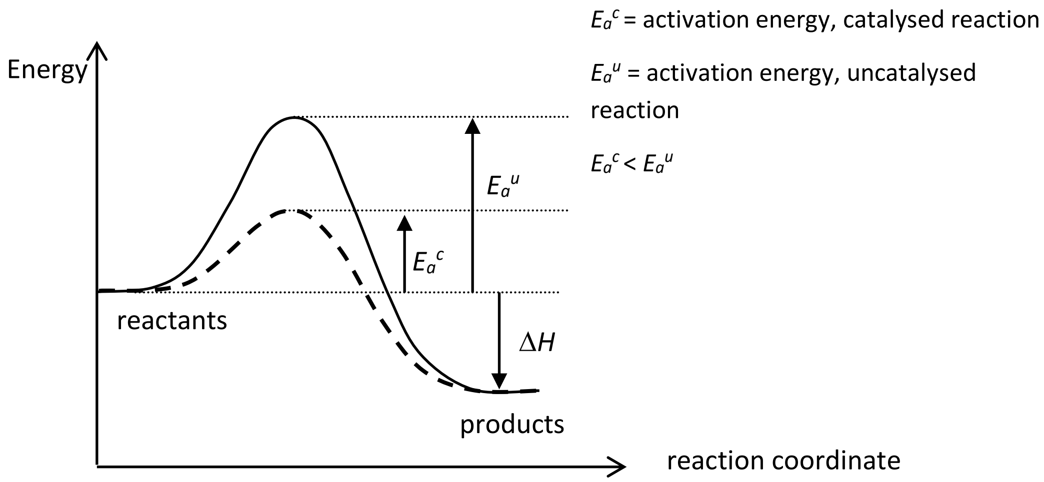
$PCl_3(l) + 3H_2O(l) \rightarrow H_3PO_3(aq) + 3HCl(aq)$

(b) **(i)** Define catalyst.

A catalyst is a substance that increases the rate of a chemical reaction but remains chemically unchanged at the end of the reaction.

(ii) With the help of an energy diagram, explain the function of a catalyst in a reaction. **[3]**

A catalyst provides an alternative reaction pathway with a lower activation energy as shown in the graph. As a result, more reactant particles possess the energy required for an effective collision and the frequency of effective collisions increases. The rate of reaction is thus increased.



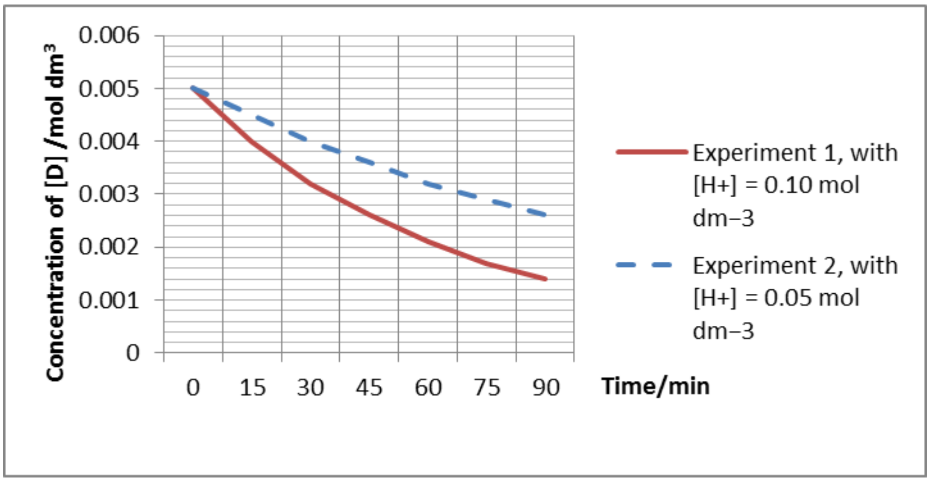
* Boltzmann distribution graph is accepted too.

(c) The rate of hydration of organic compound **D** was followed by measuring the concentration of **D** after fixed time intervals. Two experiments were carried out, starting with different concentrations of H^+. The following results were obtained.

Time/min	Experiment 1, with $[H^+] = 0.10$ mol dm^{-3}	Experiment 2, with $[H^+] = 0.05$ mol dm^{-3}
0	0.0050	0 0050
15	0.0040	0.0045
30	0.0032	0.0040

45	0.0026	0.0036
60	0.0021	0.0032
75	0.0017	0.0029
90	0.0014	0.0026

(i) Using the same axes, plot graphs of [**D**] against time for two experiments



(ii) Using the graphs, determine the order of reaction with respect to H^+ and **D**.

Find the initial rate for experiment 1 and 2

When $[H^+]$ is halved, the initial rate of reaction is halved, order of reaction with respect to $[H^+]$ is one.

From expt 1, the graph shows a constant half-life of 50min**.**

Order of reaction with respect to [D] is one.

(iii) Hence, write the rate equation for this reaction.

Rate = $k[D][H^+]$

(iv) Calculate the rate constant, including its units. **[10]**

$$k = \frac{7.4x10^{-5}}{(0.10)(0.0050)} = 0.148 \text{ mol}^{-1} \text{ dm}^3 \text{ min}^{-1}$$

[Total: 20]

8. An organic compound **A** have the molecular formula of $C_6H_{11}Cl$.

When **A** is heated with KOH in ethanol, a mixture of two isomers **B** and **C**, with the formula C_6H_{10}, are produced.

B reacts with hot acidified $KMnO_4$ to give **D**, C_5H_8O, as well as CO_2 and water. Compound **D** does not give a silver mirror with Tollen's reagent, but gives an orange precipitate when added to 2,4-DNPH.

C reacts with hot acidified $KMnO_4$ to give a straight chain compound **E**, $C_6H_{10}O_3$ only. Compound **E** does not give a silver mirror with Tollen's reagent, but gives an orange precipitate when added to 2,4-DNPH. Compound **E** also gives a yellow precipitate and salt **F**, $C_5H_6O_4Na_2$, when mixed with I_2 in NaOH(aq).

(a) By providing appropriate explanation with regards to the reactions, deduce the structure of **A**, **B**, **C**, **D**, **E** and **F**. **[13]**

From formula of **A**, it either contains a C=C or it is a cyclic chloroalkane

A undergo elimination reaction to form **B** and **C.**

B and **C** are alkenes/ contain alkene group(s).

D does not reacts with Tollen's reagent but undergo condensation with 2,4-DNPH

D is a ketone

B undergoes (vigorous) oxidation to give **D** and CO_2 is produced.

Hence, **D** contains one $R_2C{=}CH_2$ group/ **D** has a terminal alkene.

Hence, **A** is a cyclic chloroalkane since **B** only contain a single C=C

C undergoes (vigorous) oxidation to give **E**

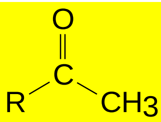
C contains only one C=C and **E** is a straight chain. Hence, **C** has a ring structure

E does not reacts with Tollen's reagent but undergo condensation with 2,4-DNPH

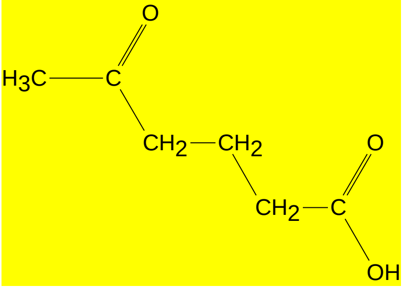
E has a ketone group

Since **E** contains 3 Oxygen, and **C** contains only one C=C, it have a carboxylic group

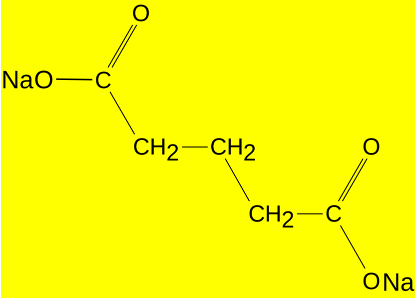
E is oxidised by I_2 in NaOH / gives a positive iodoform test

Hence the **E** contains  group.

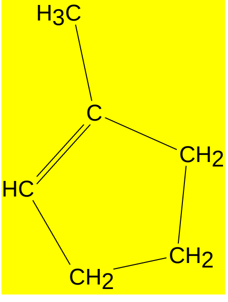
Since **E** is straight chain, it must have the structure of:



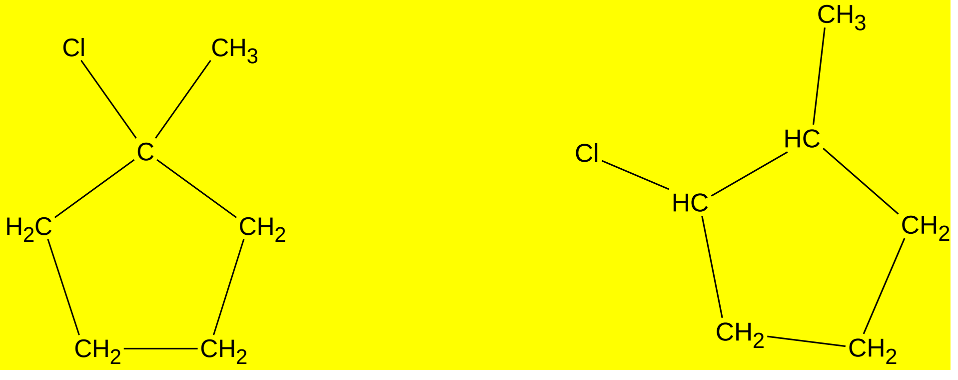
Hence, **F** is:



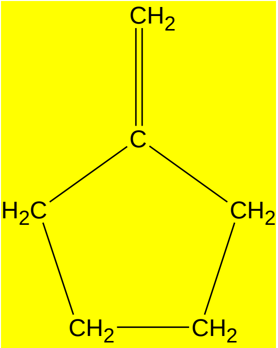
To produce **E**, alkene **C** must be:



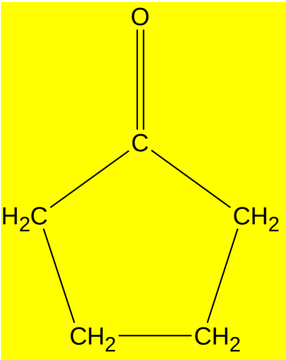
A can either be: or:



But only the first structure can give **B**:



Which can be oxidized to give **D**:

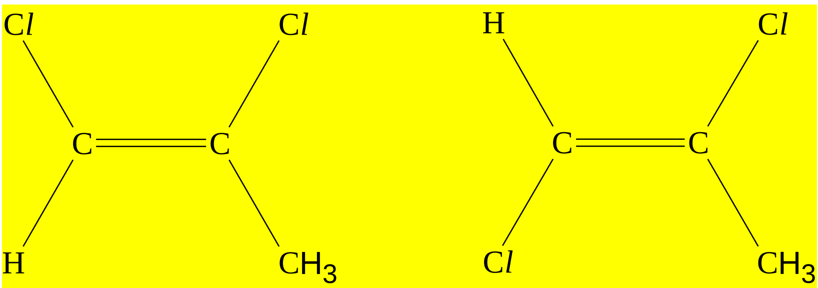


and CO_2 and water

(b) During the synthesis of compound **A**, an organic compound, 1,2–dichloropropene, is produced as a byproduct.

1,2–dichloropropene can exist in two different isomeric form.

(i) Draw the structure of both isomers.



cis1,2–dichloropropene trans 1,2–dichloropropene

(ii) State and explain which isomeric form is expected to have a higher boiling point. **[5]**

The cis isomer would have a higher boiling point.

The cis isomer has a (overall) net dipole moment while the trans isomer do not. More energy is required to overcome the stronger permanent dipole–permanent dipole between the cis isomer compared to the weaker instantaneous dipole–induced dipole of the trans isomer.

(c) Comparing 2–chloropropanoic acid and 3–chloropropanoic acid: state and **[2]**

explain which compound is the stronger acid.

2–chloropropanoic acid is the stronger acid. The electron withdrawing chlorine group in 2–chloropropanoic acid is closer to the carboxyl group and is better able to stablise the $-COO^{-}$ group/ Conjugate base compared to 3–chloropropanoic acid.

[Total 20]